

EM 659 : Advanced Lean Systems - Course Syllabus

Muhammad Ahmed

mahmed6@emich.edu

Office: 111 Sill Hall

Office Hours: Monday 11:00 - 5:00PM

Phone: 2487033544

Fax: Skype= drmsmed

Textbook(s)

- 1) **Required:** Due to nature of the course we decided to combine two books into one using McGraw Hill's CREATE. The book is available online as eBook. You can order and download from <https://create.mcgraw-hill.com/shop/> and search ISBN 9781121430891 or EM659 Advanced Lean Systems. If you have any issue with ordering or downloading

the book please call _ [800-962-9342](tel:800-962-9342) and select option "student" and then Tech Support to seek solutions.

Also you can visit the Operation Management Textbook website at [Stevenson Student's Web](#)

- 2) **Required: The Toyota Way** -- Liker J
ISBN: 0-07-139231-9 McGraw Hill N.Y, (2004)
- 3) **Required: EM659 Advanced Lean Systems** -- Multiple Authors
ISBN 9781121430891 McGraw Hills , 2012

Course Description A class in advanced lean practices as applied in manufacturing and the office. Focus will be on the advanced practical aspects on implementing and maintaining enterprise systems capable of continuous improvement. In-depth treatment will be given value stream analysis as defined by customers, lean logistics, lean cost accounting, zero defect quality practices, and the integration of human resource practices within the enterprise strategy.

Course Objectives EM659 Advanced Lean Systems explores advanced lean practices as applied in manufacturing. This is the third course in Lean Track for all Graduate Engineering Management students at Eastern Michigan University. The course explores lean production systems and focus on the practical aspects on implementing and maintaining production systems capable of continuous improvement. This course will help students to understand the mathematical models utilized in Lean production systems and acts as a first preparatory course in SME Bronze Lean Certification.

At the completion of this course you should be able to:

- A. Trace the evolution of enterprise systems from craft to mass to lean forms.
- B. Articulate the advantages and disadvantages of the lean form of enterprise

organization both in a vertically integrated and a de-verticalized global business environment.

- C. Demonstrate understanding of the general elements of lean and their inter-relationship.
- D. Describe the general social and technical issues of implementing lean enterprise in both brownfield and greenfield environments
- F. Provide examples of lean quality techniques and interventions which can lead to zero defects
- G. Describe how the lean business strategy integrates human and technical resources
- H. Develop a rationale for the integration of total productive maintenance (TPM) into lean enterprise.
- I. Be able to describe and conduct a lean transformation analysis in a real-world setting
- E. Demonstrate a basic mastery of conventional mass enterprise cost accounting as well as lean cost accounting and activity-based systems.

Course Policies and Procedures

1. Students are responsible for monitoring messages on opening page several times a week.
2. Students are held responsible for reading and other assignments, tests, information, and materials dispensed during the course.

3. No time extensions will be given for assignments unless agreed upon by the instructor. Late assignments and tests are subject to a reduced grade. Assignments submitted within a week will receive a maximum of one-half the value. Assignments submitted more than a week late will receive a grade of zero.

4. "I"(s) will be given only in accordance with Eastern Michigan University's guidelines.

5. Feedback concerning course requirements, assignments, tests, or content is welcome and encouraged.

6. Students experiencing difficulty with course content should seek a private consultation with the instructor to discuss remediation of deficiencies.

7. Academic Honesty: Please review the university policy on academic honesty and plagiarism at <http://www.emich.edu/sjs/conductcode.html>

Eastern Michigan University Conduct Code and Judicial Structure for Students and Student Organizations

I.A.11. Academic dishonesty

1. Students are not to engage in any form of academic dishonesty including, but not limited to, plagiarism, alteration of records, substitution of another's work representing it as the student's own, and knowingly assisting another student in engaging in any such activity. For purposes of this section, plagiarism is defined as the knowing use, without appropriate approval, of published materials, expressions, or works of another with intent to represent the material(s) as one's own.

Grading Policy Grades will be determined after a statistical analysis of the class performance on all assignments. Letter grades will be granted based on the criteria listed in the Graduate Catalog.

10	Unit Exams
35	Individual Assignments
25	Team Assignment
15	Threaded Response
15	Final Exam

Grading Scale

100 to 93% = A 92.99 to 90% = A- 89.99 to 87% = B+ 86.99 to 83% = B 82.99 to 80% = B- 79.99 to 77% = C+ 76.99 to 73% = C 72.99 to 70% = C- (no D grades, based on Grad School requirements) 69.99 and below = E

Final course grades will be based primarily (but not necessarily only) on the quantitative scale shown above--but are subject to adjustment based on the judgment of your professor in consideration of qualitative factors, including academic dishonesty and displays of other unethical, unprofessional, or uncivil behaviors that are unbecoming of a good engineering manager and graduate student. (You must to be prepared for the reality of professional life in business and industry--there is usually more to getting a good performance evaluation from your boss and peers than just production of work, other factors include: (a) personal professional skills, (b) professional communication skills, and (c) ethics and emotional intelligence.) (a) (b) and (c) above are of vital importance to becoming successful as an Engineering Manager, and why they are three of our EM Program Student Learning Objectives. We expect to see them demonstrated by students in our classes.